

Amazon T Level Project

RESEARCH AND PROPOSAL

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Proposal

For our website, we took the outstanding array of t-level opportunities that are offered all around the UK and at Amazon specifically and gave them the platform to shine, helping young students to be able to easily find out the important information they need to discover what course best suits them and their career path or aspirations, attracting more students towards the amazing opportunity that a t-level can offer. These t-levels allow students to gain valuable real-world experience that can help them excel in the real world and we wanted to create a website that allowed this message to shine, being accessible for any student to navigate and discover any information they would like to, no matter what additional learning or accessibility needs they may have.

Business Context

The website would be created for students, parents and teachers to access and view information on T-Levels in Amazon to create career pathways for the younger generation and for Amazon to recruit them. Due to Amazon's wish to attract more attention for their t-level courses, the websites and solutions we have created will allow them the opportunity to showcase these courses and spread the word of the increasing positive feedback they are creating in educational program for all students, no matter of any factors or learning disabilities.

Functional and Non-Functional Requirements

Function requirements:

- Provide information on and advice about T-Levels in general and at Amazon
- Allow Amazon to collect expression of interests from customers.
- Downloadable content
- Provide access to digital content to support customers.
- Social features

Non-functional requirements:

- Accessibility
- Security features
- Scalability

Decomposition of Problem

The problems that would need to be solved to implement the functional and non-functional requirements broken down into sections are:

- Getting information on T-levels in general and at Amazon
- Implementing downloadable content appropriately
- Providing appropriate access to customers while making sure that their access level matches only their needs and nothing more.
- Implementing social features without compromising the safety of users.
- Making sure that the website is scalable
- Making sure that the website security matches modern standards
- Creating a proper system to collect expressions of interest.

Key Performance Indicators (KPIs)

- Unique users that visited the site
- Percentage of users that registered interest
- Percentage positive feedback
- Time spent on website
- Downloadable Content
- User Demographic
- Percentage Return to website

Proposed Solution

Recommendation 1:

In this solution, we decided to take a different approach to a normal standard website and traditional theoretical learning techniques, instead forming an idea, harnessing physical learning styles, that is adapted and geared towards both students who like learning the traditional way but also those who learn better or prefer to learn by being hands on. By adapting the website to cater for these needs, more kids can easily be able to understand the information that is being presented to them, allowing them the equal opportunity to get involved with these great t-level opportunities.

The adding of game-like features also help to keep kids from the ages to 11-18 interested and engaged with the information they are ingesting as it turns it from information that they might not be interested in into a fun game that lets them learn the key aspects of the t-level they would want to join. The addition of these features keeps kids wanting to learn more, allowing t-levels that students might not know about before to have more exposure.

Recommendation 2:

The Website would be an informative website which would allow users to learn about T-Levels in general and in Amazon. Users would also be able to register and create accounts to view or download certain content as well as using those accounts to send expressions of interests to Amazon. This informatic approach allows users to be able to discover what type of t-levels are available across the UK whilst helping them to work out what career path they might want to pursue. By adding a description of what Amazon specifically have to offer, it will help to attract more students to the outstanding opportunities that they offer, allowing more students to gain the many benefits of their t-level courses.

Due to Amazon's t-levels not being as well-known as they should be by students and schools, emphasising these t-levels will not only help to shed light on what they have to offer but it will also allow students to learn what career path they would like to take from a younger age, as a result of them having the easy access to the right support and information they need. Consequently, this makes amazon's t-level programs stand out more than another organisation's and make them more popular for students to choose to help them with their careers. This could then possibly lead into career paths into amazon because of their great experience on the t-level, bringing in more experienced workers for amazon, improving both their own skills and amazon's business.

Recommended Solution based on Client and User Needs

Recommendation 1:

Platform Architecture & UX Zones

- **Living Campus:** The overarching User Experience (UX) framework for the platform, designed to simulate a lively, buzzing university campus rather than a flat, sterile Learning Management System (LMS).
- **The Quad (The Out-of-Class Social Layer):** A dynamic social feed inspired by platforms like LinkedIn and TikTok, but strictly restricted by age tier.
 - For users aged 16–18, it serves as a public repository for project wins, hackathon submissions, and text posts.
 - For users under 16, the feed is restricted to a "Read-Only" or "In-Cohort-Only" stream to demonstrate safeguarding compliance.

- **The Project Labs (The Vocational Cohort Layer):** The structured digital classrooms where actual skill development takes place. These are dedicated digital spaces assigned to specific learner groups (e.g., "Cloud Practitioner Cohort Beta"). The interface includes dedicated sidebar slots displaying the online status of Student Peers, College Tutors, and AWS Mentors.
- **The Library (The Autonomous Pathway Layer):** A quiet digital workspace where the student's individual learning plan is housed. It features a clean dashboard that displays the user's progress through three distinct milestones: exploring (ages 11–13), active courses (ages 14–16), and the career pipeline track with Amazon (ages 16–18).

Technical & Engagement Mechanics

- **Passive AI Agent:** Described as the "Siri" for learning, this is an autonomous system that quietly monitors a student's progress and builds personal context in the background. It ingests telemetry—such as completed modules, code commits, and project lab chat topics—to run Retrieval-Augmented Generation (RAG) queries that synthesize custom summaries, bespoke quizzes, or adapted learning pathways on demand.
- **80/20 Content Strategy:** The specific engagement hook utilized within The Quad's feed. It mandates that 80% of the feed consists of bite-sized, student-generated multimedia content (videos, code snippets, project demos), while the remaining 20% is dedicated to deep technical masterclasses from AWS experts.
- **Competence Gates:** A gamification mechanic utilized to unlock practical project tiers based on proven mastery rather than passive consumption (e.g., simply watching a video). This is implemented to keep users engaged without triggering "Goodhart's Law," a trap where students optimize their behaviour purely to extract points rather than actually learning.
- **Relative Milestones:** A peer motivation system that replaces traditional absolute classroom leaderboards with team-based goals or a "Personal Streak" indicator, driving engagement without introducing the shame of ranking low in a class.

Architectural Compliance & Safeguarding

- The UK's incoming "Australia-Plus" under-16 social media ban mandates a strictly tiered architecture to qualify as an exempt educational service.
- Users aged 11–15 must be restricted to read-only feeds or closed cohort spaces.

- Only users aged 16–18 can access the full public "Quad" professional networking features.
- Accessibility is a strict legal liability under the Equality Act 2010, requiring complete WCAG 2.2 AA compliance.
- The *Stephen Campbell v HSCNI* case, resulting in a £3,000 settlement, establishes firm legal precedent that inaccessible dynamic web elements are actionable discrimination.

Pedagogical Structure (The "Project Labs")

- Self-paced, isolated online courses suffer from catastrophic failure rates, with Massive Open Online Courses (MOOCs) hovering between 3.13% and 15% completion.
- A lack of social interaction is empirically the single largest barrier to digital learning success.
- The platform must utilize a synchronized cohort architecture; locking students into peer groups with dedicated mentors routinely pushes completion rates to 85% (Harvard Business School Online) or 96% (altMBA).

Gamification & Engagement Mechanics

- Gamification yields a statistically significant, large positive effect on learning environments, verified at an effect size of $g = 0.822$.
- However, absolute leaderboards cause shame and disengagement among lower-performing students.
- The architecture should instead rely on "Competence Gates" (unlocking features based on mastery) and relative personal streaks to maintain intrinsic motivation.

The Social Feed (The "Quad")

- Traditional professional networks fail to engage youth; while young adults make up 20.5% of LinkedIn's audience, 96% of Gen Z users rarely or never actively use it.
- Gen Z users abandon platforms that fail to provide an immediate "quick win".

- The feed architecture must employ an 80/20 content ratio: 80% snackable, trend-driven micro-content to algorithmically capture attention, serving as a gateway to 20% deep educational masterclasses.
- This aligns with sociological data showing 73% of Gen Z prefer micro-content over long-form text.

The Passive AI Agent (The "Library")

- Agentic AI tutors operating on a Socratic framework (guiding, not answering) are highly effective and scalable, as evidenced by systems like Khanmigo scaling to over 700,000 users.
- Because the AI agent passively profiles minors, it triggers severe liabilities under UK GDPR and the ICO Children's Code.
- The technical architecture must replicate Apple's Private Cloud Compute model to ensure data sovereignty.
- Student telemetry must be organized via a localized semantic index and processed within an isolated Amazon Bedrock Knowledge Base inside a private VPC, ensuring zero external data retention.

Recommendation 2:

The Home page:

- This is the main page and the first page that user sees when entering the website. In this page the first thing the user will be exposed to will be pictures or videos relating to Amazon and T-levels that will cycle automatically or they can be manually cycled as well and a navigation menu on the top of the page.
- They will also see an icon on the top right of the page next to the navigation menu for an option to create an account to access features that require an account
- Below that, there will be text about why the user should pick a T-Level for their course and describing the benefits of choosing a T-Level as well as the companies that amazon.
- The home page is helpful to the user because it introduces them to the benefits of a T-Level which would be something the user would be looking for and it also allows for easy navigation between pages.

The About page:

- In this page, the user will be able to find about what is a T-Level in general and learn about how they work like the things they would learn in a T-level and the text describing the T-Level would be complemented by a picture relating to T-Levels
- After that, there will be description about Amazon which would describe who they are what their values are to show the user what they are getting into, and this would also be complemented by a picture relating to Amazon.
- This page introduces the user to T-Levels and Amazon to help them familiarise themselves and let them know what they are going to find in this website

The Content page:

- The user will first see an image/graphic and then be greeted by an introduction to give context for the user.
- Next the user will be given an image and explanation of a course with a button to cycle through the t-levels that amazon offers.
- Then there is a segment to show what companies amazon owns.
- Finally, there is a section that allows the user to receive contact information relating to T-levels that amazon provides.

The Contact page:

- This is the page where the user can contact both other users and the company itself.
- Users can access the company's email and phone number which may be used to contact them.
- This company info can be used by the user to contact the company meaning it meets the user requirement of 'supporting customers' part of the brief.
- This company info is at the bottom of the page as it is a conventional layout
- Users are recommended people they may know, like LinkedIn and social media such as Facebook and Instagram. Additionally, they can view profiles they have viewed previously

- This means that amazon record data about each profile about how often and who it is viewed by meaning it meets the 'collect expressions of interest from all customers' section of the brief
- There is a search bar at the top of the page as it is a convention. This search bar can be used to search for profiles they wish to contact
- There are arrows on either side of the profiles so the user can scroll through the profiles. This is good as it means that more profiles can be displayed without taking up more space on the page.

Further additions:

- After the Core features and pages have been added, we will then focus on adding more features to the website if we have time and these would include:
 - o Separate pages for each T-Level (e.g. A separate page for Digital T-Levels)
 - o Social features which would require an account to use so users can share their opinions and experiences with T-Levels
 - o A Frequently Asked Questions page (FAQ) to help users resolve and questions or issues they have.
 - o Age verification for safety and so that people of only similar ages can interact with each other

Final Pros and cons of our recommended solutions:

Recommendation 1:

Pros	Cons
Gamified approach which causes increased retention.	Would be harder for the team to implement.
AWS uses Generative AI and is a fan of it being used in their processes/ systems.	Risks for the user is higher because due to requiring age verification for social features.
Appeals more and is better for the target age demographic	Lack of accessibility.
Stands out from other regular websites.	Harder to implement onto another server

Recommendation 2:

Pros	Cons
More of a theoretical approach to learning.	Less engaging for readers.
Simpler for our team to implement, meaning more of us can help to program the website.	Doesn't appeal to age demographic.
Easier for us and Amazon to maintain.	Doesn't utilise AI to provide extra information or guidance on the website
Easier to make more accessible for all students	Could have too much information on the pages which could cause students to lose interest or struggle to read

Final recommended solution:

There is a myriad of reasons that we ended up deciding to move on with proposal one instead of two. To begin, we recommend using our first solution because it is more engaging as it has a game aspect to it which appeals to the younger generation. And as it is more original, it allows the project to seem more unique & appealing, both to potential users & Amazon. Additionally, as it is a little harder to implement than proposal two, it allows our team to challenge ourselves, alongside upskilling. Consequently, by upskilling our team, we will gain more knowledge and experience that can help us become better programmers and problem solvers in our future careers, giving us valuable skills that employers look for whilst producing a high-quality website that helps Amazon to positively advertise their own and other t-levels across the UK.

As the world progresses alongside digital solutions, the use of generative AI is more common across all industries and individuals, making it rapidly grow into a feature that is used in most of the people's life daily. Due to this, we made sure to include this very important feature to make sure that our website is easy to access and allows students to gain even more information than they would be able to without it whilst also helping them to also gain skills in using it and how to create the best outcome possible, learning key skills of how to harness and use this AI tool in the correct and productive way.

Using a gamified approach to teaching the students about the t-levels helps to accommodate the students who prefer to learn in a more physical and hands-on way, allowing them to have the equal chance of learning more about the courses and career they would like to pursue, making the website more accessible for more students. This helps to make our website stand out from other websites with similar information as it appeals more to the age demographic that will be using the website. By doing this, we can keep the students engaged with the information we would like them to see and will help to draw in more students to follow and join the various courses that are available to them.

Risk Mitigation

- Risk Mitigation is important for many reasons:
 - o Protection against digital threats stopping unauthorised access and data breaches
 - o Ensures a stable company reputation as data breaches and other threats are avoided
 - o Compliance with laws and legislations such as GDPR
 - o Decision making is enhanced as risk mitigation guides a company to safer and more effective solutions
- Not attempting to add features outside of our capabilities to prevent wastage of time and effort.
- Risks should be systematically identified with consideration of the likelihood and severity of the risk.
- This information should then be used to plan ways to avoid these errors using a Risk Management Plan

Security Practices

- Secure and robust code
- Testing and reviewing the product before launch to make sure that risks are minimised.
- Input validation
- Making sure that users have the minimum and only needed permissions that would be needed to use the website.
- Use Firewalls to inspect incoming traffic to prevent and block incoming attacks towards the website to prevent and data breaches or hacks.

Regulatory and Legal Guidelines

- Equality act 2010 means it's illegal to discriminate against people based on a disability and we could face repercussions if we fail to meet the requirements of this law
- Accessibility regulations 2018 means you must meet the WCAG 2.2 AA accessibility guidelines. This means that we must have features such as alt text for images or tags for screen readers.
- GDPR or General Data Protection Regulation is also applied here as we would be storing user's credentials and as such, we would need to make sure that data we store is secure.

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